

CUONG PHAM (Phạm Anh Cường)

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Machine Learning researcher focused on **efficiency** and **optimization** for deep learning.

Research interests: [continual learning](#), [parameter-efficient fine-tuning of LLMs](#), [collaborative/federated learning](#), [optimization](#)

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), UAE 2024 - 2026
Master of Science in Machine Learning GPA: 3.72/4.0 (~ 94%) [Transcript]

- Selected courses: Mathematics for AI, Advanced Machine Learning, Probabilistic & Statistical Inference, Optimization.
- Thesis: “Learning in the Null Space: Small Singular Values for Continual Learning”.
- Supervisors: [Prof. Samuel Horváth](#) and [Prof. Praneeth Vepakomma](#).

VNU University of Engineering and Technology (VNU-UET), Vietnam 2019 - 2023
Bachelor in Information Technology (Honors Program) GPA: 3.71/4.0 (8.78/10) [Transcript]

- Graduated with High Distinction, top 5% of CS Department.
- Thesis: “Building a Vietnamese image captioning dataset and developing a visual-text transformation model for generating image captions”.
- Supervisors: [Prof. Quang-Thuy Ha](#), [Dr. Van-Quang Nguyen \(RIKEN AIP\)](#), [MS. Thi-Hong Vuong](#).

PUBLICATIONS

(P = preprint, C = conference, O = ongoing paper)

- [O.6] **Cuong Anh Pham**, Praneeth Vepakomma, Samuel Horváth: (Ongoing project on efficient continual fine-tuning)
- [C.5] **Cuong Anh Pham**, Praneeth Vepakomma, Samuel Horváth: “*Learning in the Null Space: Small Singular Values for Continual Learning*”
(CPAL '26 - The Third Conference on Parsimony and Learning, **Oral Presentation**).
- [P.4] Farshed Abdukhakimov, **Cuong Anh Pham**, Samuel Horváth, Martin Takáč, Slavomir Hanzely: “*Polyak Stepsize: Estimating Optimal Functional Values Without Parameters or Prior Knowledge*”
(preprint 2508.17288, under-review draft).
- [P.3] Quang P.M. Pham, Khoi T. N. Nguyen, Nhi H. Doan, **Cuong A. Pham**, Qinbo Sun, Weimin Qi, Kentaro Inui, Dezhen Song: “*SmallPlan: Leverage Small Language Models for Sequential Path Planning with Simulation-Powered, LLM-Guided Distillation*”
(preprint 2505.00831).
- [P.2] **Anh-Cuong Pham**, Van-Quang Nguyen, Thi-Hong Vuong, Quang-Thuy Ha: “*KTVIC: A Vietnamese Image Captioning Dataset on the Life Domain*”
(preprint 2401.08100, short version of Bachelor’s thesis).
- [C.1] Quynh-Trang Pham Thi, **Anh-Cuong Pham**, Ngoc-Huyen Ngo and Duc-Trong Le: “*Memory-Based Method using Prototype Augmentation for Continual Relation Extraction*”
(IEEE RIVF '22).

EXPERIENCE (Research)

MBZUAI, Abu Dhabi, UAE

Research Engineer

06/2026 - Present

- Research focuses on efficient collaborative learning and privacy-aware training under data-constrained settings.
Supervisor: [Prof. Praneeth Vepakomma](#)

Graduate Research Assistant

10/2024 - 05/2026

- Research on orthogonality-based low-rank updates for mitigating catastrophic forgetting in continual learning and parameter-efficient fine-tuning of foundation models (**Thesis Project**, coauthor paper [C.5], ongoing work [O.6]).
Supervisors: [Prof. Samuel Horváth](#) and [Prof. Praneeth Vepakomma](#)
- Research on adaptive and parameter-free stepsize for stochastic optimization algorithms based on Polyak stepsize with twin iterates (coauthor paper [P.4]).
Supervisors: [Prof. Samuel Horváth](#) and [Prof. Martin Takáč](#)
- Study federated optimization and federated fine-tuning methods, reproduce and analyze techniques such as Fed-SB and LoRA-SB (Summer Intern Topic).
Supervisors: [Prof. Praneeth Vepakomma](#) and [Prof. Samuel Horváth](#)

- Research on the application of small language models (SLMs) for robotic path planning through simulation-guided distillation from large models (LLMs) (coauthor paper [P.3]).

DSKT Laboratory at VNU-UET, Hanoi, Vietnam

09/2021 - 09/2023

Student Research Intern

- Research on vision-language learning for low-resource datasets and lightweight Transformer-based captioning models (**Thesis Project**, coauthor paper [P.2]).
- Research on memory-based continual relation extraction using prototype augmentation (coauthor paper [C.1]).

EXPERIENCE (Industry)

Viettel Networks, Hanoi, Vietnam

10/2023 - 07/2024

Data Scientist

- Research and develop computer vision solutions for telecommunication network maintenance.
- Analyze large-scale telecommunication datasets and develop predictive models for network optimization.

GHTK Joint Stock Company, Hanoi, Vietnam

05/2022 - 08/2023

R&D Engineer (~ AI Engineer), with 3 months as R&D Engineer Intern

- Research and develop an email spam filtering model for internal company applications.

INVITED TALKS

“**Learning in the Null Space: Small Singular Values for Continual Learning**”

- CPAL 2026, hosted by ELLIS Tübingen & MPI-IS & Tübingen AI Center (Tübingen, Germany) 2026
- RelationalML Group at CISPA, hosted by Prof. Rebekka Burkholz (online) 2026

HONORS & AWARDS

Oral Presentation at CPAL, Tübingen, Germany	2026
Full scholarship for 2 years of the Master of Science program at MBZUAI	2024
(3×) Third Prize (ranked 5th, 6th, 5th) in Linear Algebra, VNU-UET Mathematical Olympiad	2021, 2022, 2023
Yamada Scholarship (6 students/year) at VNU-UET	2022
Excellent Scholarship for top highest GPA from VNU-UET (rank 5th)	2021
Attended Vietnam Mathematical Olympiad for high school students (ranked among top 250/600 students)	2019

CONFERENCES, SCHOOLS & SEMINARS ATTENDED

Efficient Machine Learning Seminar at MBZUAI (Abu Dhabi, UAE)	2025-present
(CPAL '26) The Third Conference on Parsimony and Learning (Tübingen, Germany)	2026
(MLWS '26) MBZUAI Machine Learning Winter School (Abu Dhabi, UAE)	2026
(ICOMP '25) The International Conference on Computational Optimization (Abu Dhabi, UAE)	2025
(ASCOMP '25) The AI Autumn School of Computational Optimization (Abu Dhabi, UAE)	2025

SKILLS

Programming: Python, C/C++

Tools: Microsoft Office, L^AT_EX, Git, Linux.

Frameworks: PyTorch, Scikit-learn.

Language: English (IELTS 7.0), Vietnamese (native).

REFERENCES

Dr. Samuel Horvath, Assistant Professor at MBZUAI (UAE): (samuel.horvath@mbzuai.ac.ae)
Dr. Praneeth Vepakomma, Assistant Professor at MBZUAI & Visiting Professor at MIT IDSS: (vepakom@mit.edu)